

● EASY

● ADVANCED MATH

Advanced Math

07

10 questions · ~15 min

Q1

ADVANCED MATH

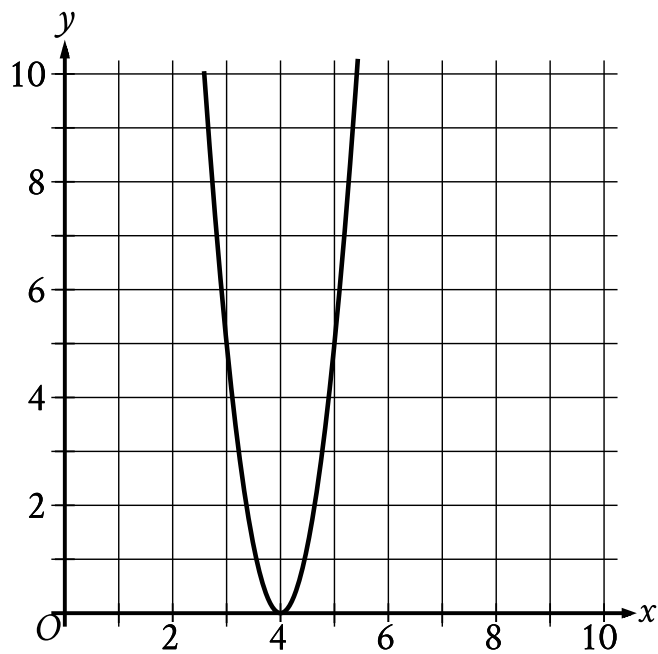
Which expression is equivalent to $4(x^2 + 6)$?

A. $4x^2 + 24$

B. $4x^2 + 10$

C. $4x^2 + 6$

D. $4x^2 - 2$



- The parabola opens upward.
- The vertex is at point (4 comma 0).
- The parabola passes through the following points:
 - (3 comma 5)
 - (4 comma 0)
 - (5 comma 5)

What is the x-intercept of the graph shown?

- A. -5,0
- B. 5,0
- C. -4,0
- D. 4,0

Q3

ADVANCED MATH

$$q - 29r = s$$

The given equation relates the positive numbers q , r , and s . Which equation correctly expresses q in terms of r and s ?

- A. $q = s - 29r$
- B. $q = s + 29r$
- C. $q = 29rs$
- D. $q = -\frac{s}{29r}$

Q4

ADVANCED MATH

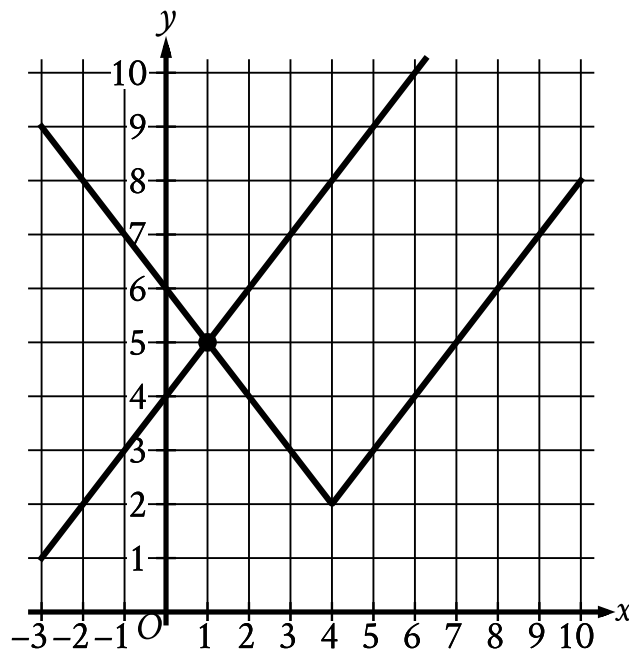
Which expression is equivalent to $16x + 15$?

- A. $16x + 31$
- B. $16x + 240$
- C. $16x + 1$
- D. $16x + 15$

$$f(x) = (x + 0.25x)(50 - x)$$

The function f is defined above. What is the value of $f(20)$?

- A. 250
- B. 500
- C. 750
- D. 2,000



- For the absolute value function:
 - Moving from left to right:
 - The function slants sharply down to the point (4 comma 2).
 - The function then slants sharply up.
 - The function passes through the following points:
 - (1 comma 5)
 - (4 comma 2)
 - (6 comma 4)
- For the linear function:
 - The function slants sharply up from left to right.
 - The function passes through the following points:
 - (0 comma 4)
 - (1 comma 5)

The graph of a system of an absolute value function and a linear function is shown. What is the solution x, y to this system of two equations?

A. -1,5

B. 0,4

C. 1,5

D. 4,2

Q7

ADVANCED MATH

Which expression is equivalent to $x^{\frac{1}{14}}$, where $x > 0$?

A. $\frac{1}{14} \cdot x$

B. $\sqrt[14]{x}$

C. $14 \cdot x$

D. x^{14}

The function $f(x) = 240,000(1.22)^x$ gives a company's predicted annual revenue, in dollars, x years after the company started selling jewelry online, where $0 < x \leq 10$. What is the best interpretation of the statement " $f(5)$ is approximately equal to 648,650" in this context?

- A. 5 years after the company started selling jewelry online, its predicted annual revenue is approximately 648,650 dollars.
- B. 5 years after the company started selling jewelry online, its predicted annual revenue will have increased by a total of approximately 648,650 dollars.
- C. When the company's predicted annual revenue is approximately 648,650 dollars, it is 5 times the predicted annual revenue for the previous year.
- D. When the company's predicted annual revenue is approximately 648,650 dollars, it is 5% greater than the predicted annual revenue for the previous year.

$$x + 45 = 48$$

What is the positive solution to the given equation?

- A. 3
- B. 48
- C. 93
- D. 96

$$5x + 15$$

Which of the following is equivalent to the given expression?

- A. $5(x + 3)$
- B. $5(x + 10)$
- C. $5(x + 15)$
- D. $5(x + 20)$